Module 5 — Assembly Language Machine Problems

NASM 32-bit | Procedures | Conditional Selection | User Input | ASCII Art

Two interactive console applications • Intermediate difficulty • Recommended time: 2–3 hours per problem

General Programming Requirements

Develop both applications using NASM x86 32-bit assembly on Linux with int 0x80. Programs must compile and run without errors.

Requirement	Minimum
Procedures using CALL and RET	5
User input operations	3
Conditional selections	5
Menu options	4
ASCII art designs	3
Menu repetition or loop	1
Input validation and exit option	Required

Machine Problem 1 — Zodiac Horoscope and Fortune Teller

Problem Statement

Create an interactive Zodiac Horoscope and Fortune Teller System. The user enters a name, birth month and birth day. The program identifies the zodiac sign and displays a fictional horoscope according to a selected category. Separate major tasks into procedures and use conditional branches for date and menu decisions.

1. Welcome Screen

```
=======================================
       THE MYSTIC ZODIAC ORACLE
=======================================
              *       *
           *     /\       *
                /  \
          *    /____\    *
                |  |
            *   |__|   *
       DISCOVER YOUR DESTINY!
=======================================
```

2. User Information

```
Enter your name: Maria
Enter birth month (1-12): 8
Enter birth day (1-31): 15

Welcome, Maria!
```

Validate month and day against actual month lengths. February 29 may be rejected because no birth year is collected.

3. Zodiac Identification

Sign	Date range
Capricorn	Dec 22–Jan 19
Aquarius	Jan 20–Feb 18
Pisces	Feb 19–Mar 20
Aries	Mar 21–Apr 19
Taurus	Apr 20–May 20
Gemini	May 21–Jun 20
Cancer	Jun 21–Jul 22
Leo	Jul 23–Aug 22
Virgo	Aug 23–Sep 22
Libra	Sep 23–Oct 22
Scorpio	Oct 23–Nov 21
Sagittarius	Nov 22–Dec 21

Display the sign, element, symbol, personality description, and simplified ASCII art corresponding to the selected sign.

```
========================================
         YOUR ZODIAC SIGN
========================================
              /\_/\
             ( o.o )
              > ^ <
Zodiac Sign: LEO
Element: Fire
Symbol: Lion
Personality: Confident, creative, courageous.
```

4. Horoscope Menu

```
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
Enter your choice:
```

Choices 1–3 display category-specific fictional predictions; choice 4 displays a lucky number and color; choice 5 shows zodiac details and art; choice 6 exits. Invalid selections must display an error and return to the menu.

5. Sample Interaction

```
Enter your name: Maria
Enter birth month: 8
Enter birth day: 15
Hello, Maria!
Your Zodiac Sign: LEO
Your Element: FIRE
Choice: 2
CAREER HOROSCOPE
Leo, your creativity will help you
solve a difficult problem today.
Stay focused and confident!
Return to menu? (Y/N): Y
```

Required Procedures

Procedure	Responsibility
display_welcome	Display welcome ASCII art
get_user_info	Read name and birth date
validate_birthdate	Check date validity
determine_zodiac	Identify zodiac sign
display_zodiac_art	Display sign-specific ASCII art
display_menu	Display horoscope choices
show_horoscope	Select and show prediction
exit_program	Terminate program

Required branching: CMP, JE/JNE for menus, and JL/JG/JLE/JGE for date ranges; include a menu loop.

Test Cases

Test	Input	Expected result
1	March 21	Aries
2	April 19	Aries
3	April 20	Taurus
4	June 15	Gemini
5	July 23	Leo
6	August 15	Leo
7	December 25	Capricorn
8	February 30	Invalid date
9	Month 13	Invalid month
10	Menu choice 9	Invalid selection

Test Case 1:



```
====================================
      THE MYSTIC ZODIAC ORACLE
====================================
                *
               ***
              *****
             *******
            *********
              | |
              | |
             ========
         DISCOVER YOUR DESTINY!
====================================

Enter your name: Vlad
Enter birth month (1-12): 3
Enter birth day (1-31): 21
Welcome, Vlad

====================================
         YOUR ZODIAC SIGN
====================================

Zodiac Sign: ARIES
Element: FIRE
Symbol: Ram
Personality: Courageous, determined, confident.
   (\\_/)
   (o.o)
   (> <)  (Ram)
```

Test Case 2:


THE MYSTIC ZODIAC ORACLE
DISCOVER YOUR DESTINY!
Enter your name: Vlad
Enter birth month (1-12): 4
Enter birth day (1-31): 19
Welcome, Vlad
YOUR ZODIAC SIGN
Zodiac Sign: ARIES
Element: FIRE
Symbol: Ram
Personality: Courageous, determined, confident.
(_/)
(o.o)
(> <) (Ram)

Test Case 3:


THE MYSTIC ZODIAC ORACLE
DISCOVER YOUR DESTINY!
Enter your name: Vlad
Enter birth month (1-12): 4
Enter birth day (1-31): 20
Welcome, Vlad
YOUR ZODIAC SIGN
Zodiac Sign: TAURUS
Element: EARTH
Symbol: Bull
Personality: Reliable, patient, practical.
(__)
(oo)
/-||-\\ (Bull)

Test Case 4:



Test Case 5:



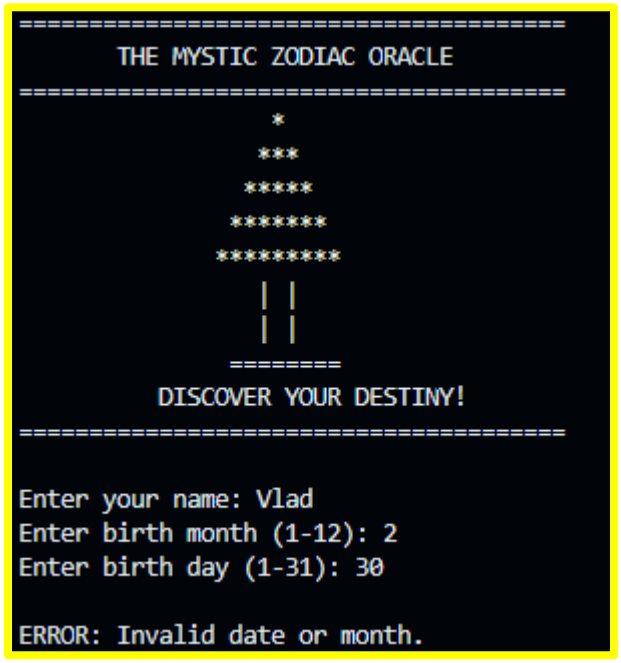
Test Case 6:



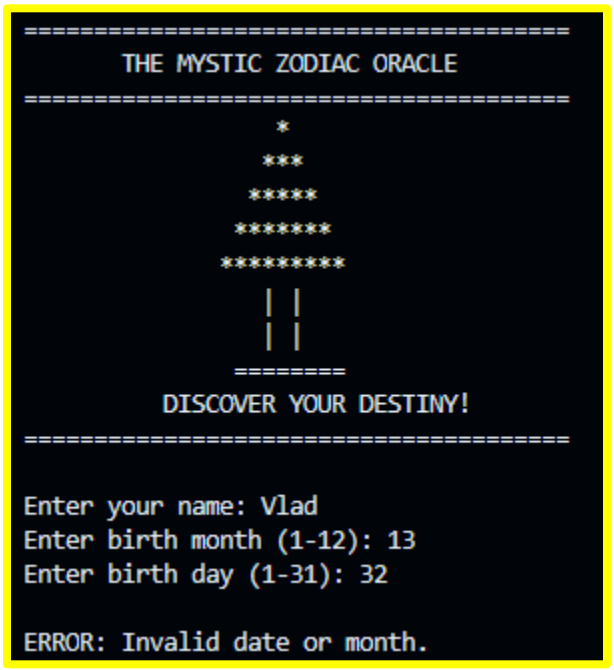
Test Case 7:



Test Case 8:



Test Case 9:

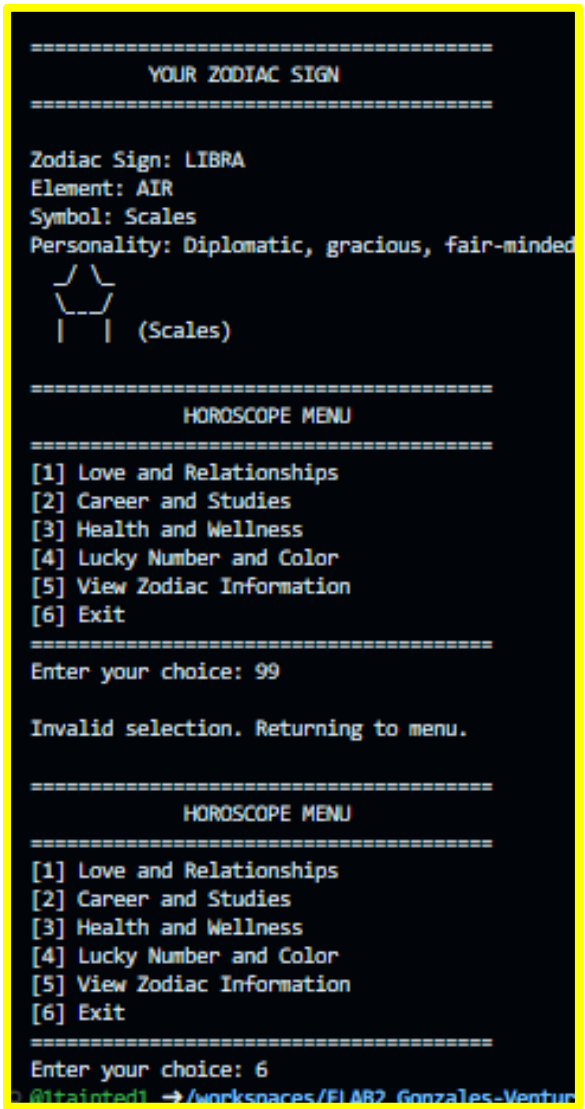


Test Case 10:



```
========================================
       THE MYSTIC ZODIAC ORACLE
========================================
                *
               ***
              *****
             *******
            *********
               | |
               | |
             =======
          DISCOVER YOUR DESTINY!
========================================

Enter your name: Vlad
Enter birth month (1-12): 9
Enter birth day (1-31): 30
Welcome, Vlad
```



```
========================================
          YOUR ZODIAC SIGN
========================================

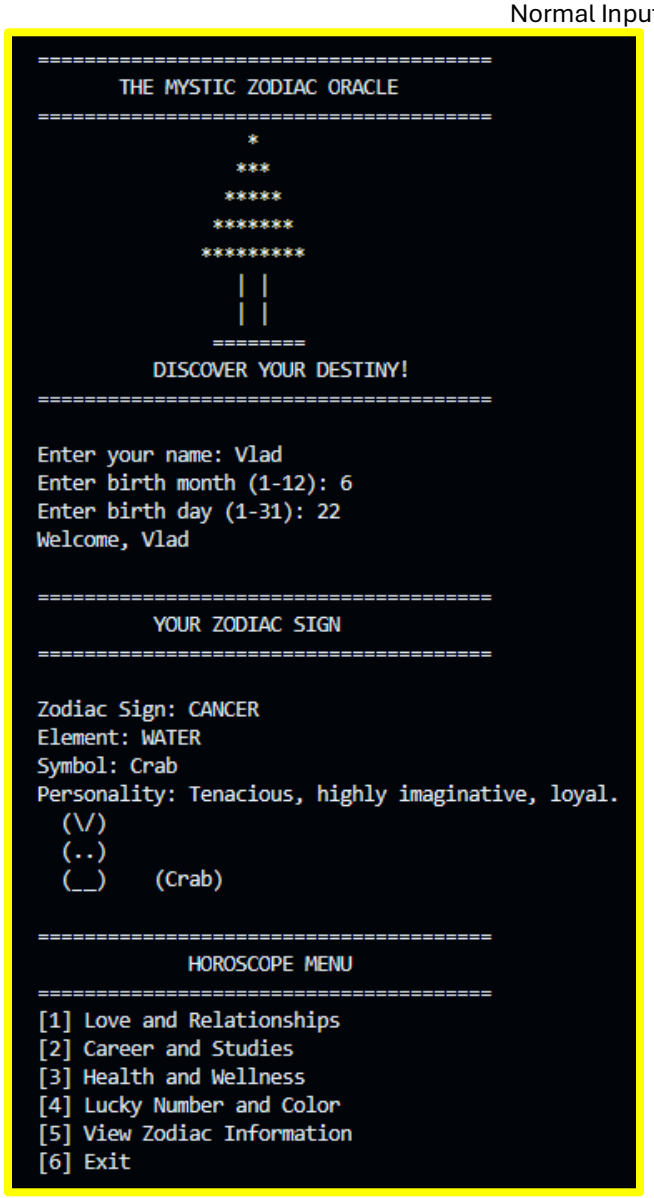
Zodiac Sign: LIBRA
Element: AIR
Symbol: Scales
Personality: Diplomatic, gracious, fair-minded
  _/ \_
  \___/
  |   |  (Scales)

========================================
             HOROSCOPE MENU
========================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
========================================
Enter your choice: 99

Invalid selection. Returning to menu.

========================================
             HOROSCOPE MENU
========================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
========================================
Enter your choice: 6
@1tainted1 ➜ /workspaces/FLAB2_Gonzales-Ventur
```

Normal Inputs:



```
=====================================
      THE MYSTIC ZODIAC ORACLE
=====================================
                *
               ***
              *****
             *******
            *********
              | |
              | |
             ========
         DISCOVER YOUR DESTINY!
=====================================

Enter your name: Vlad
Enter birth month (1-12): 6
Enter birth day (1-31): 22
Welcome, Vlad
```

```
=====================================
Enter your choice: 1

LOVE HOROSCOPE:
A sudden spark will ignite an old connection.
Keep your heart open to unexpected surprises!

Return to menu? (Y/N): y
```

```
=====================================
            HOROSCOPE MENU
=====================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
=====================================
Enter your choice: 2

CAREER HOROSCOPE:
Your creativity will help you solve a difficult problem today.
Stay focused and confident!

Return to menu? (Y/N): y

=====================================
            HOROSCOPE MENU
=====================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
=====================================
Enter your choice: 3
```

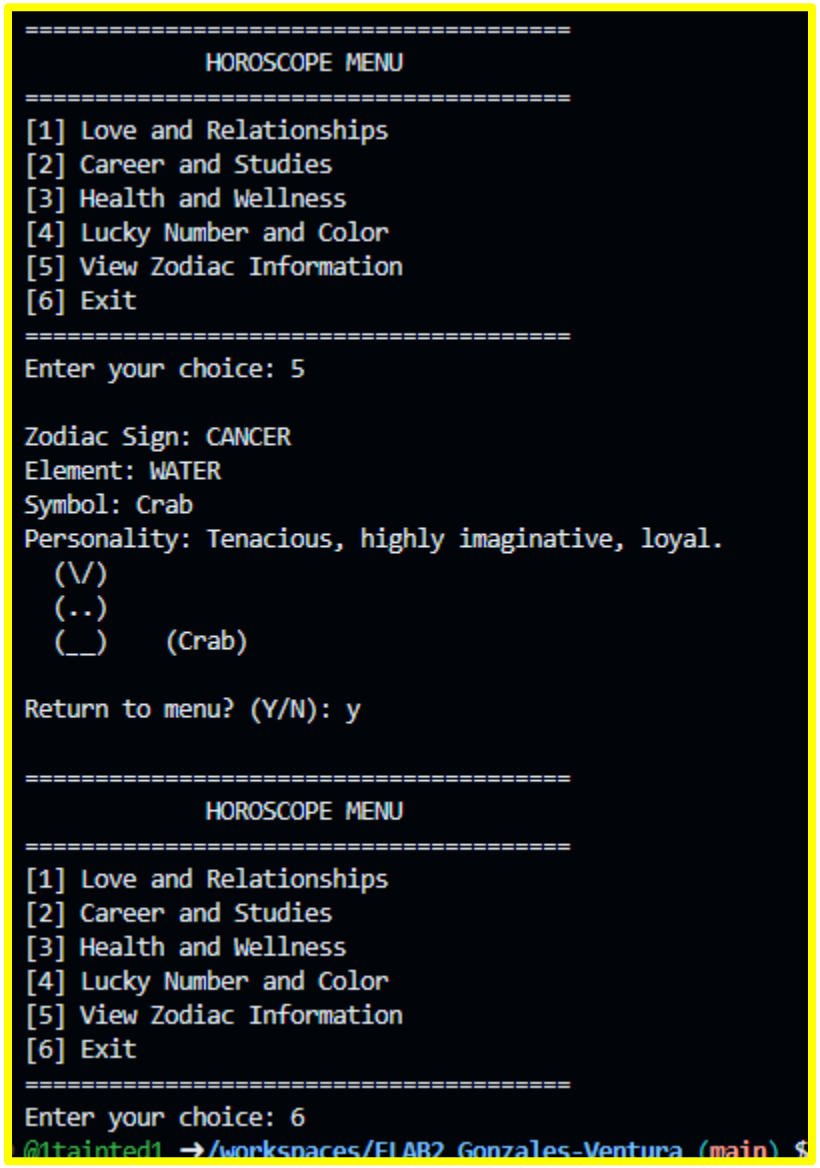
```
HEALTH HOROSCOPE:
Your energy levels are high. Good time to start a new routine.
Remember to hydrate and rest well.

Return to menu? (Y/N): y

=====================================
            HOROSCOPE MENU
=====================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
=====================================
Enter your choice: 4

LUCKY NUMBER AND COLOR:
Lucky Number: 7
Lucky Color: Royal Blue

Return to menu? (Y/N): y
```


==========
HOROSCOPE MENU
==========
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
==========
Enter your choice: 5
Zodiac Sign: CANCER
Element: WATER
Symbol: Crab
Personality: Tenacious, highly imaginative, loyal.
(\/)
(..)
(__) (Crab)
Return to menu? (Y/N): y
==========
HOROSCOPE MENU
==========
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
==========
Enter your choice: 6

Machine Problem 2 — Dungeon Adventure and Survival Game

Problem Statement

Create a text-based Dungeon Adventure and Survival Game. The player selects a class, explores three rooms, battles enemies, collects treasure, and makes decisions affecting survival. Use procedures, conditional selections, arithmetic, loops, user input, and ASCII art.

1. Welcome Screen

```
=======================================
       DUNGEON OF THE LOST KING
=======================================
               |>>>>
               |
          _  _|_  _
         |;|_|;|_|;|
         \.    .  /
          \:  .  /
           ||:   |
           ||:.  |
           ||:  .|
       ENTER IF YOU DARE!
=======================================
```

2. Player Registration

```
Enter your hero name: Arthur
Choose your character:
[1] Warrior
[2] Mage
[3] Archer
Enter choice: 1
Welcome, Warrior Arthur!
```

Character	Health	Attack	Potions
Warrior	100	20	2
Mage	70	30	3
Archer	85	25	2

3. Main Menu

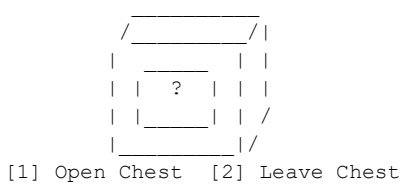
```
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
```

4. Dungeon Rooms and Gameplay

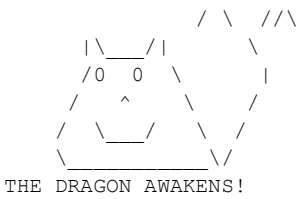
Room 1 — Goblin Chamber: Goblin health = 30. Display enemy ASCII art and offer Attack, Defend, Use Potion, or Run.

```
          ,      ,
         /(.-""-.)\
     |\  \/      \/  /|
     | \ / =.  .= \ / |
      \  \   ^    /  /
       \  '-----'  /
A GOBLIN APPEARS!
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
```

Room 2 — Treasure Chamber: The player chooses to open the chest (gain 50 gold) or leave it (gain nothing).



Room 3 — Dragon's Lair: Dragon health = 60. Show dragon ASCII art and the same combat menu as the goblin.



5. Combat and Outcome Rules

Attack: subtract player attack points from enemy health. Defend: reduce the next enemy attack by 5 (minimum damage 0). Potion: restore 20 health up to starting maximum. Run: leave encounter and return to menu without advancing. A surviving goblin deals 10 damage per turn; a surviving dragon deals 15. After Attack or Defend, the living enemy retaliates. A player loses at health 0 or less; victory occurs after defeating the dragon.

```
VICTORY!
CONGRATULATIONS, ARTHUR!
You defeated the dragon and escaped!
Final Gold: 50
Remaining Health: 35
YOU ARE A HERO!

GAME OVER
Your hero has fallen.
The dungeon remains undefeated.
BETTER LUCK NEXT TIME!
```

Required Procedures

Procedure	Responsibility
display_welcome	Show title and ASCII art
get_player_info	Read name and class
initialize_character	Assign stats
display_main_menu	Show options
explore_dungeon	Control room progression
battle_enemy	Handle combat selections
use_potion	Restore health
display_status	Show statistics
display_inventory	Show gold and potions
display_game_result	Show victory or defeat
exit_program	Terminate program

Validate class selection, menus, potion count, health boundaries, room progression, and win/loss conditions. Never allow negative displayed health or potions below zero.

Test Cases

Test	Scenario	Expected result
1	Select Warrior	Health 100, Attack 20
2	Select Mage	Health 70, Attack 30
3	Select Archer	Health 85, Attack 25
4	Select class 9	Invalid selection
5	Warrior attacks Goblin twice	Goblin defeated
6	Potion at full health	Health capped; potion consumed
7	Potion count 0	Error; no health change
8	Open treasure	Gain 50 gold
9	Health reaches 0	Game Over
10	Dragon health reaches 0	Victory screen

TEST 1: Select Warrior - Health 100, Attack 20

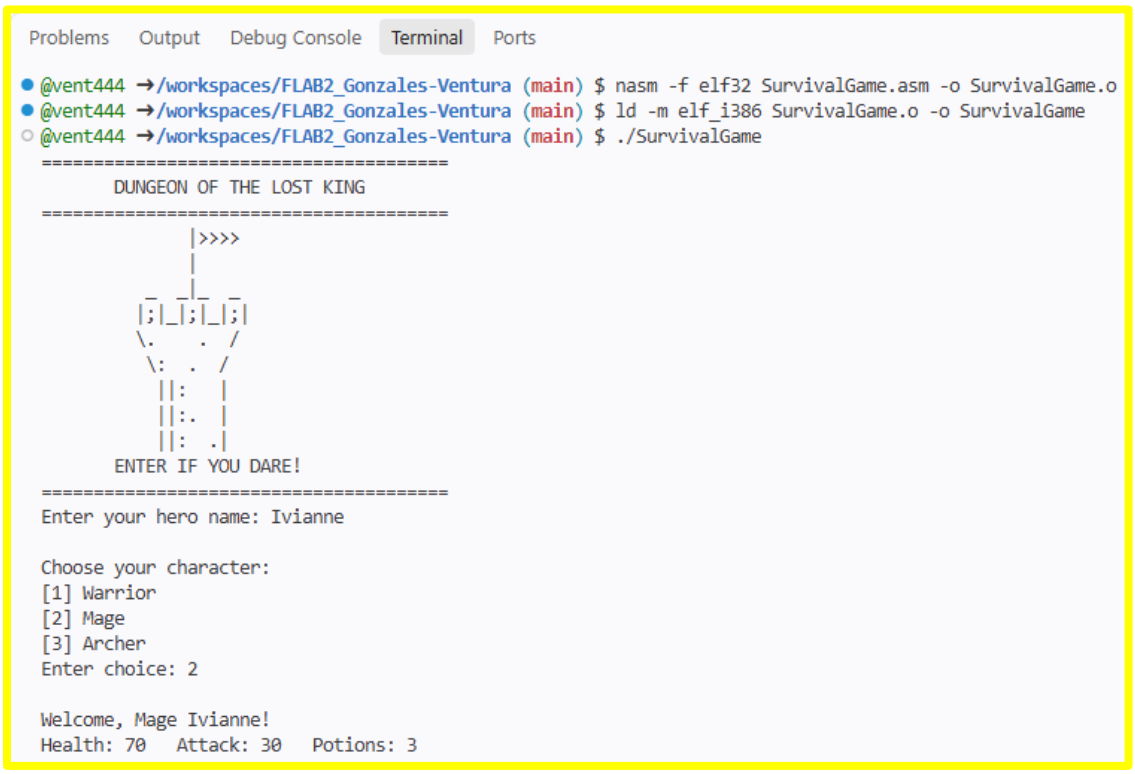
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ nasm -f elf32 SurvivalGame.asm -o SurvivalGame.o
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ld -m elf_i386 SurvivalGame.o -o SurvivalGame
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
======================================
       DUNGEON OF THE LOST KING
======================================
              |>>>>
              |
          _  _|_  _
         |;|_|;|_|;|
         \.    .  /
          \:  .  /
           ||:   |
           ||:.  |
           ||:  .|
       ENTER IF YOU DARE!
======================================
Enter your hero name: Vladimir

Choose your character:
[1] Warrior
[2] Mage
[3] Archer
Enter choice: 1

Welcome, Warrior Vladimir !
Health: 100   Attack: 20   Potions: 2
```

TEST 2: Select Mage - Health 70, Attack 30



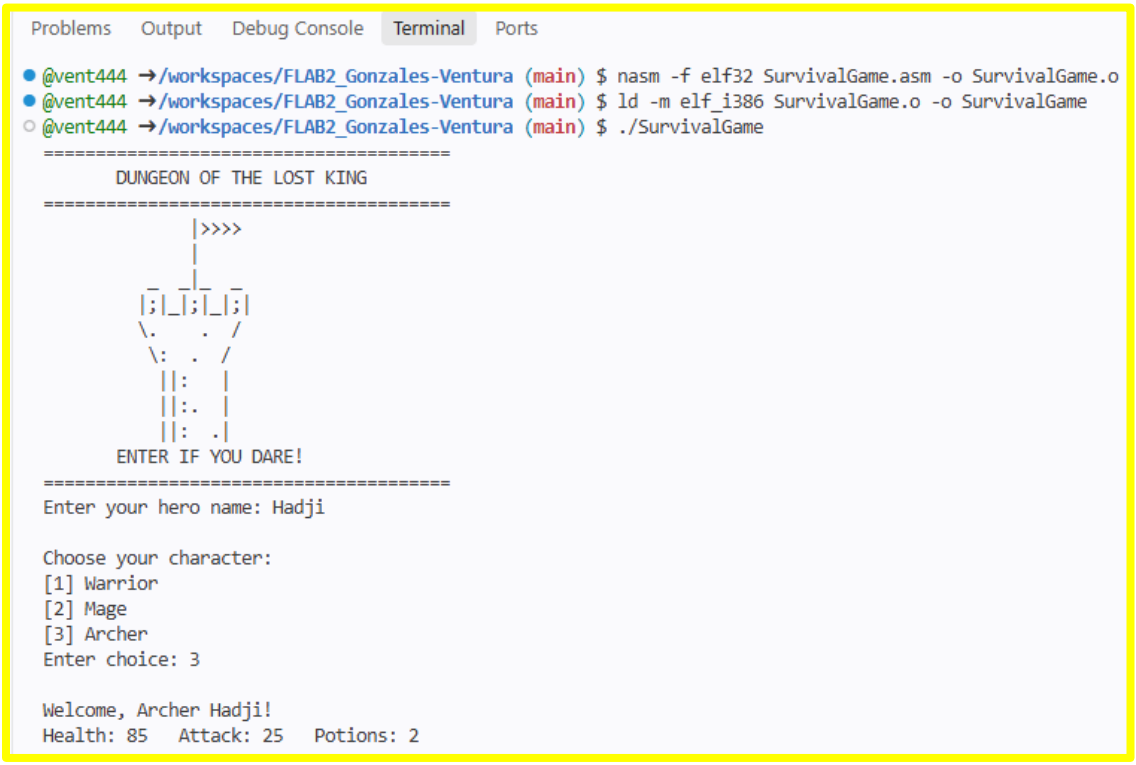
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ nasm -f elf32 SurvivalGame.asm -o SurvivalGame.o
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ld -m elf_i386 SurvivalGame.o -o SurvivalGame
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
========================================
       DUNGEON OF THE LOST KING
========================================
              |>>>>
              |
          _  _|_  _
         |;|_|;|_|;|
         \.    .  /
          \:  .  /
           ||:   |
           ||:.  |
           ||:  .|
       ENTER IF YOU DARE!
========================================
Enter your hero name: Ivianne

Choose your character:
[1] Warrior
[2] Mage
[3] Archer
Enter choice: 2

Welcome, Mage Ivianne!
Health: 70   Attack: 30   Potions: 3
```

TEST 3: Select Archer - Health 85, Attack 25



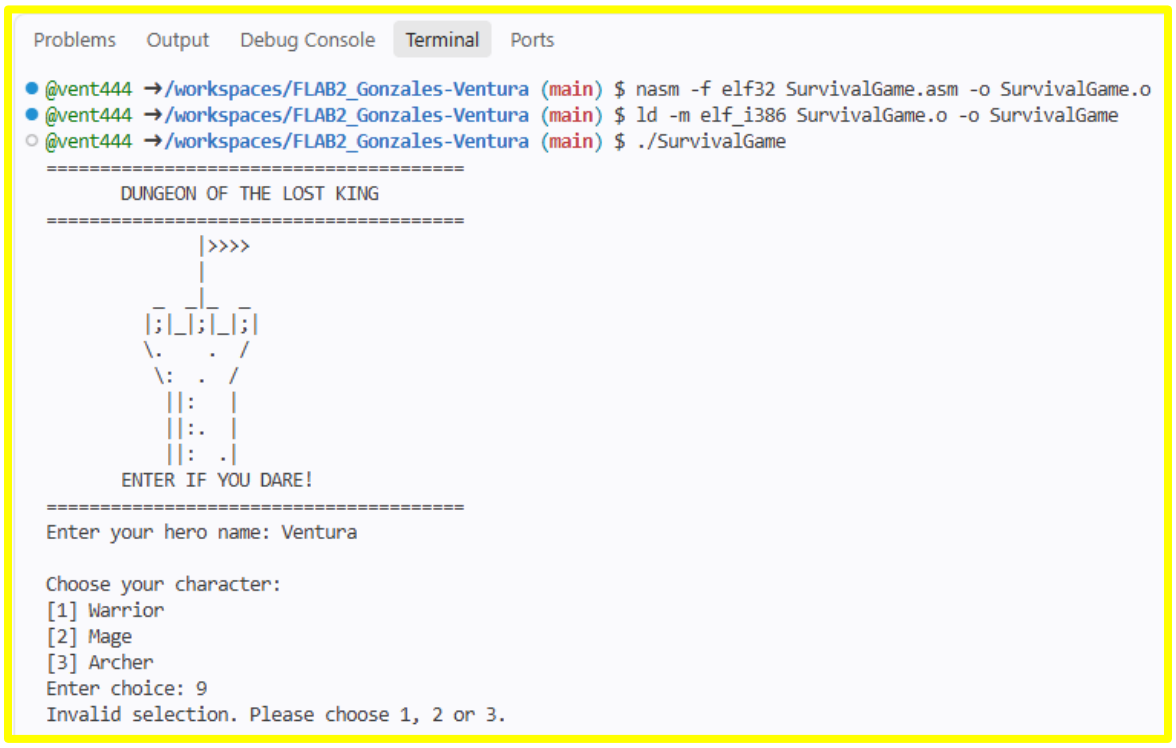
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ nasm -f elf32 SurvivalGame.asm -o SurvivalGame.o
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ld -m elf_i386 SurvivalGame.o -o SurvivalGame
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
========================================
       DUNGEON OF THE LOST KING
========================================
              |>>>>
              |
          _  _|_  _
         |;|_|;|_|;|
         \.    .  /
          \:  .  /
           ||:   |
           ||:.  |
           ||:  .|
       ENTER IF YOU DARE!
========================================
Enter your hero name: Hadji

Choose your character:
[1] Warrior
[2] Mage
[3] Archer
Enter choice: 3

Welcome, Archer Hadji!
Health: 85   Attack: 25   Potions: 2
```

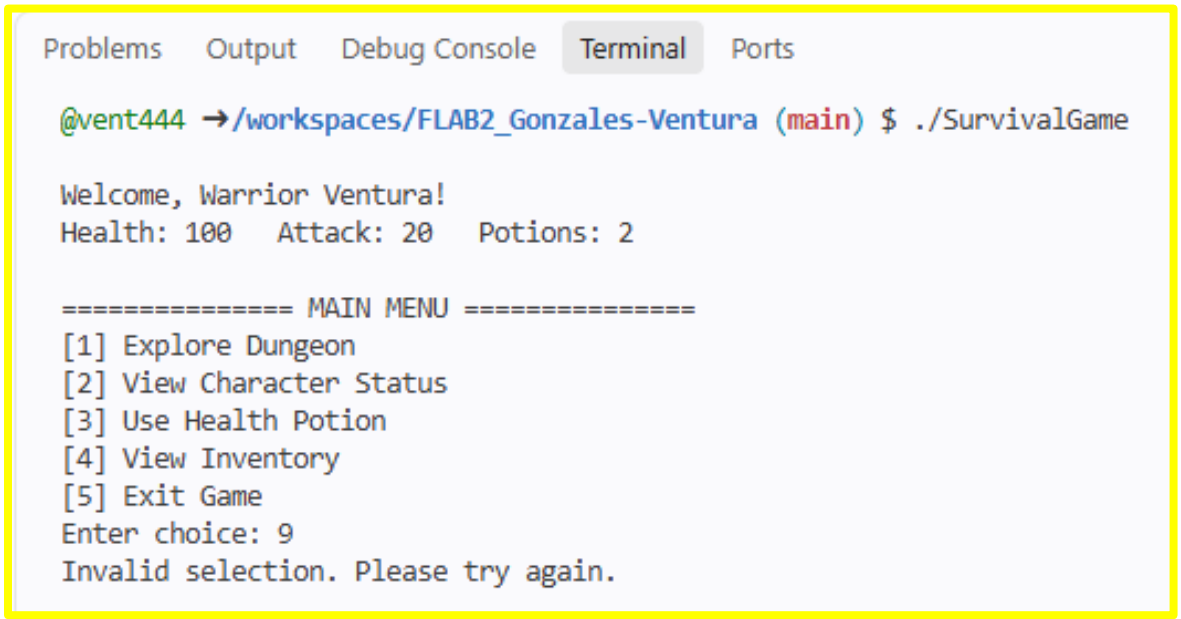
TEST 4: Select class 9 - Invalid selection



```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ nasm -f elf32 SurvivalGame.asm -o SurvivalGame.o
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ld -m elf_i386 SurvivalGame.o -o SurvivalGame
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
======================================
       DUNGEON OF THE LOST KING
======================================
             |>>>>
             |
         _   _|_  _
        |;|_|;|_|;|
        \.    .  /
         \:  .  /
          ||:   |
          ||:.  |
          ||:  .|
       ENTER IF YOU DARE!
======================================
Enter your hero name: Ventura

Choose your character:
[1] Warrior
[2] Mage
[3] Archer
Enter choice: 9
Invalid selection. Please choose 1, 2 or 3.
```



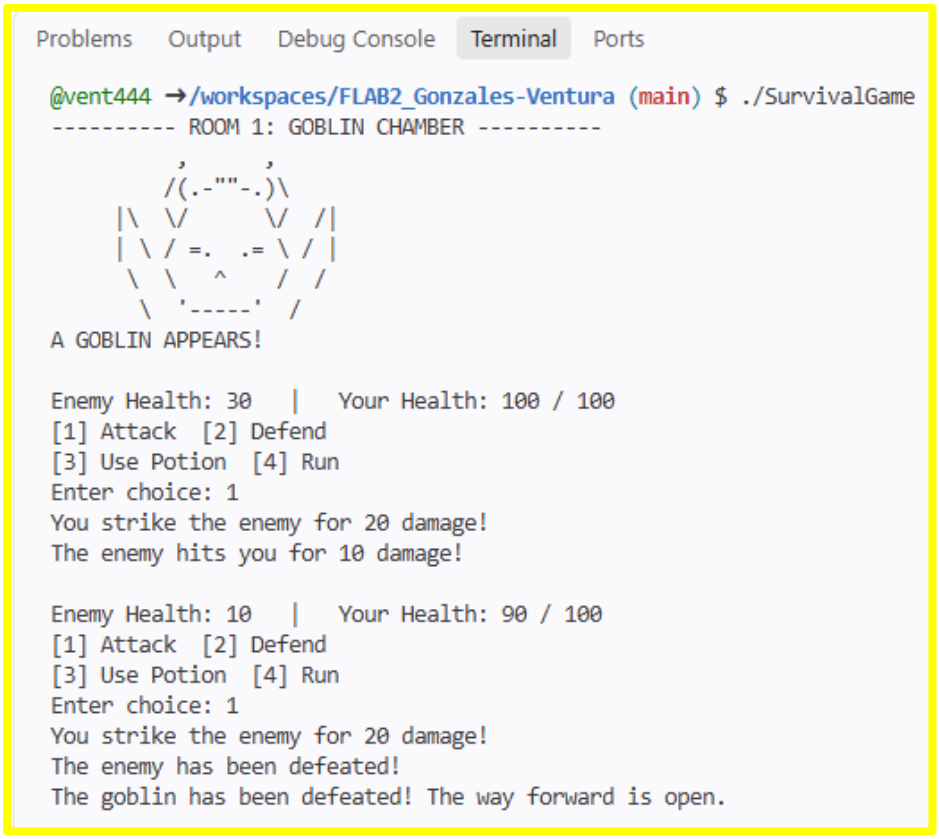
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame

Welcome, Warrior Ventura!
Health: 100   Attack: 20   Potions: 2

=============== MAIN MENU ===============
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 9
Invalid selection. Please try again.
```

TEST 5: Warrior attacks Goblin twice - Goblin defeated



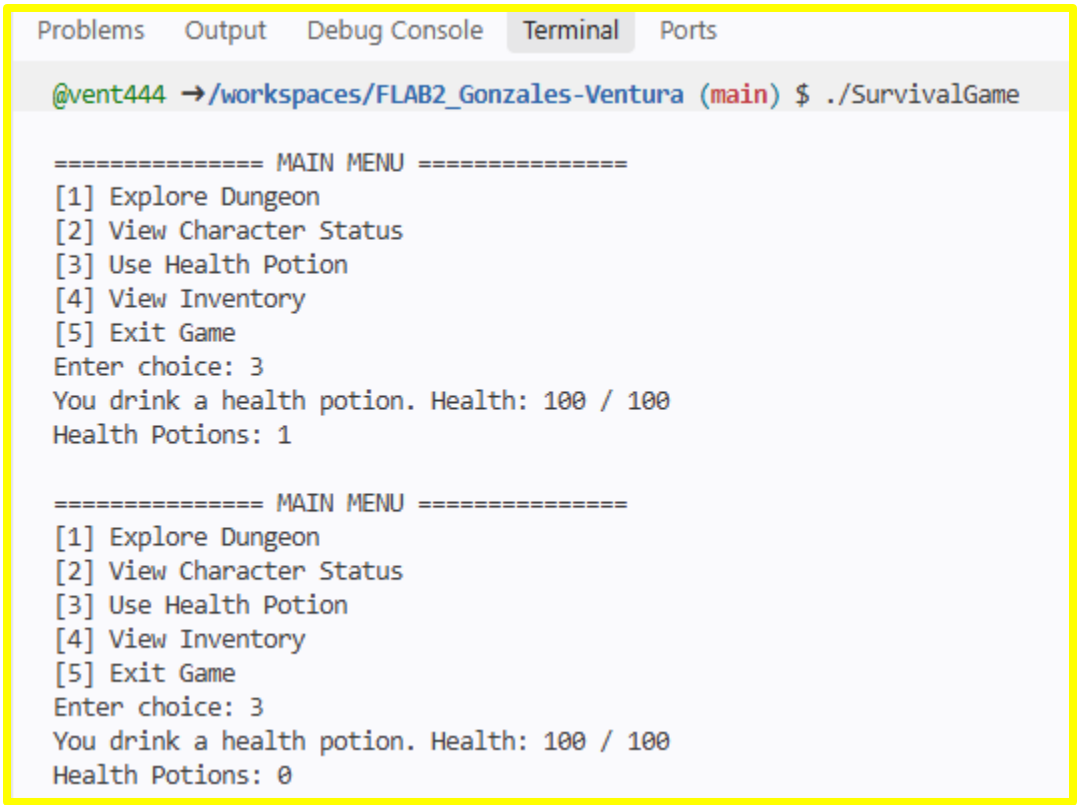
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
---------- ROOM 1: GOBLIN CHAMBER ----------
          ,      ,
         /(.-""-.)\
     |\  \/      \/  /|
     | \ / =.  .= \ / |
      \  \   ^    /  /
       \  '-----'  /
A GOBLIN APPEARS!

Enemy Health: 30   |   Your Health: 100 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 1
You strike the enemy for 20 damage!
The enemy hits you for 10 damage!

Enemy Health: 10   |   Your Health: 90 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 1
You strike the enemy for 20 damage!
The enemy has been defeated!
The goblin has been defeated! The way forward is open.
```

TEST 6: Potion at full health - Health capped; potion consumed



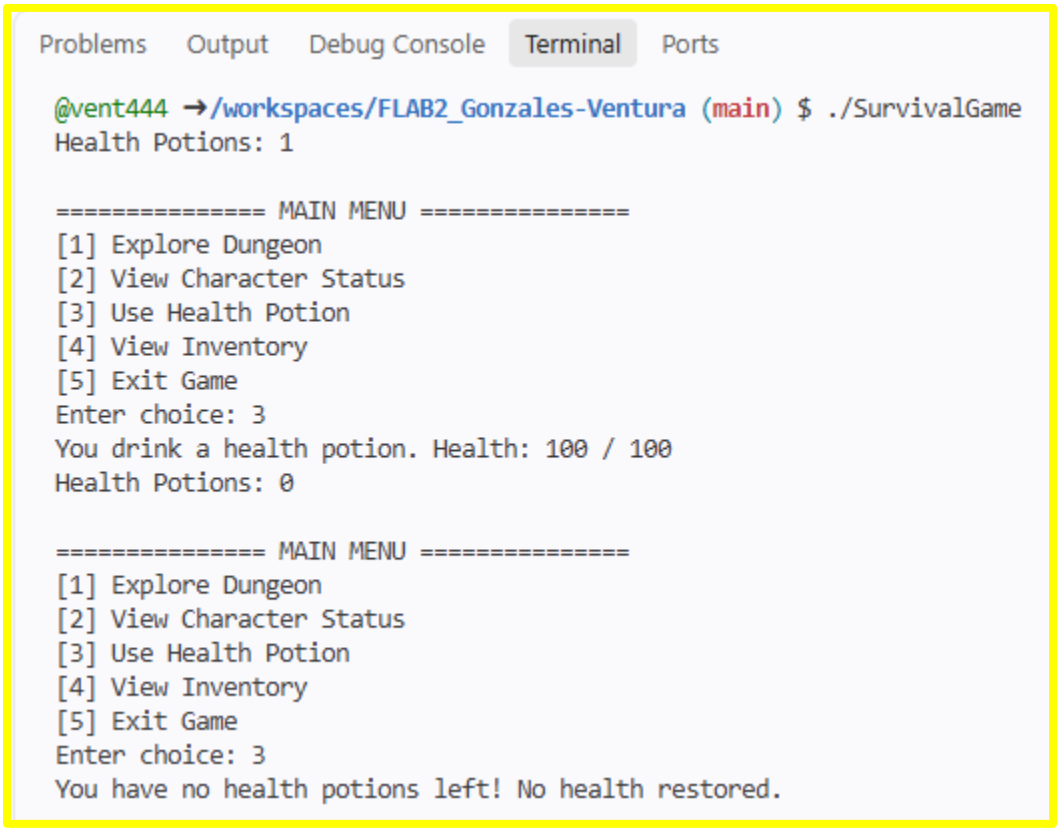
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame

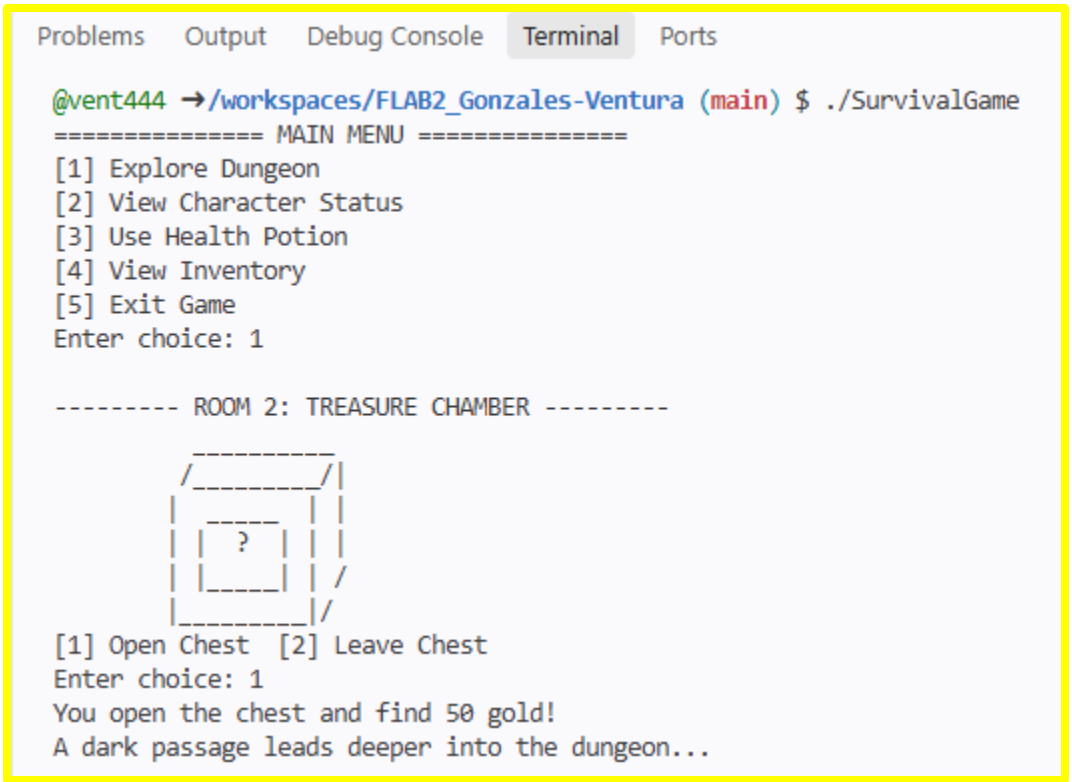
=============== MAIN MENU ===============
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 3
You drink a health potion. Health: 100 / 100
Health Potions: 1

=============== MAIN MENU ===============
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 3
You drink a health potion. Health: 100 / 100
Health Potions: 0
```

TEST 7: Potion count 0 - Error; no health change



TEST 8: Open treasure - Gain 50 gold



TEST 9: Health reaches 0 - Game Over

Problems Output Debug Console Terminal Ports

```
@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
============== MAIN MENU ==============
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 1

---------- ROOM 3: DRAGON'S LAIR -----------
               / \  //\
     |\___/|      \
     /0  0  \      |
    /   ^    \    /
   /  \___/   \  /
   \___________\/
THE DRAGON AWAKENS!

Enemy Health: 60   |   Your Health: 100 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 90 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 80 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 70 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 60 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 50 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!
```

```
Enemy Health: 60   |   Your Health: 40 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 30 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

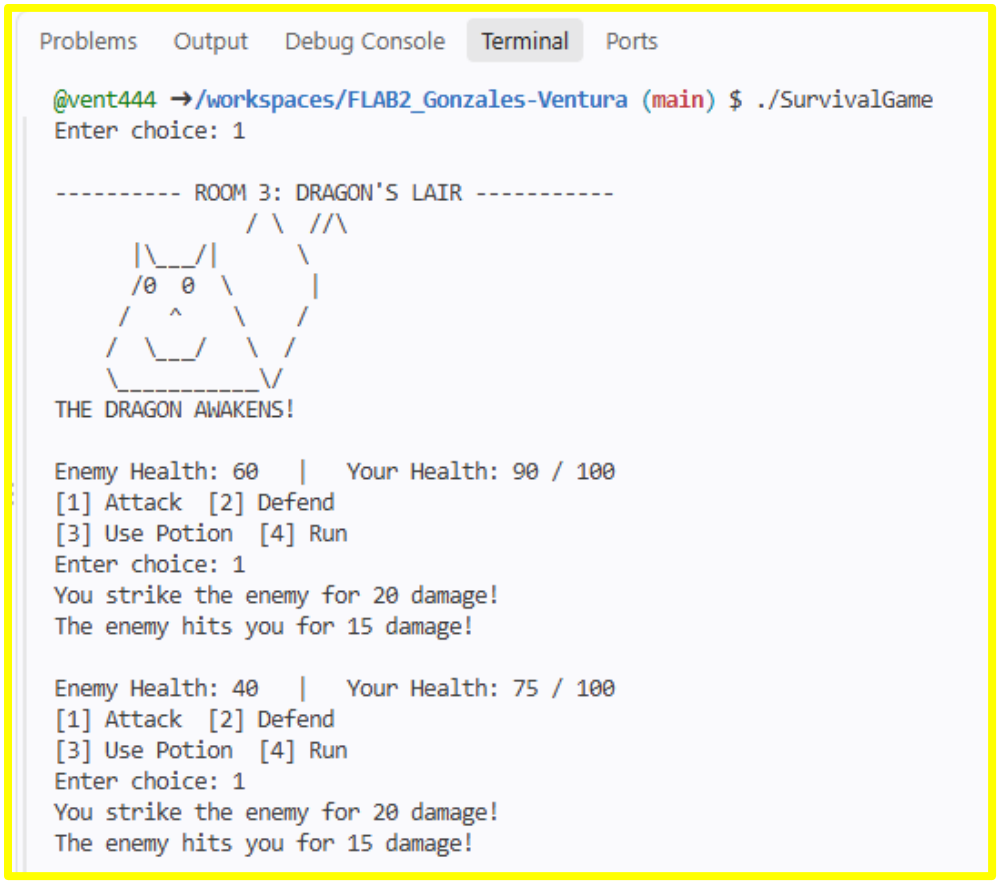
Enemy Health: 60   |   Your Health: 20 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!

Enemy Health: 60   |   Your Health: 10 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 2
You raise your guard and brace for the next attack!
The enemy hits you for 10 damage!
```

```
========================================
                GAME OVER
========================================
Your hero has fallen.
The dungeon remains undefeated.
BETTER LUCK NEXT TIME!

Thanks for playing Dungeon of the Lost King. Goodbye!
@vent444 →/workspaces/FLAB2_Gonzales-Ventura (main) $
```

TEST 10: Dragon health reaches 0 - Victory screen



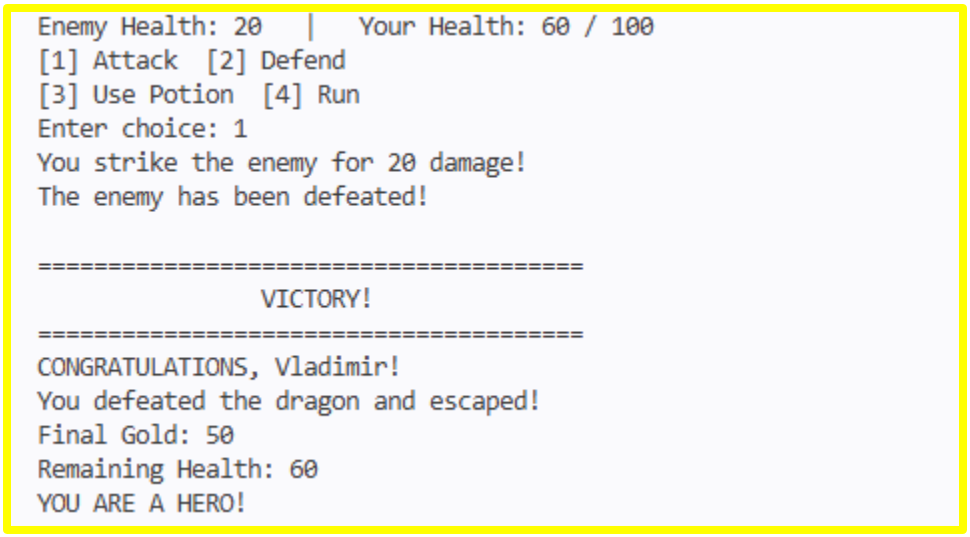
```
Problems  Output  Debug Console  Terminal  Ports

@vent444 ➜/workspaces/FLAB2_Gonzales-Ventura (main) $ ./SurvivalGame
Enter choice: 1

---------- ROOM 3: DRAGON'S LAIR -----------
               / \  //\
      |\___/|      \
      /0  0  \      |
     /   ^    \    /
    /  \___/   \  /
    \___________\/
THE DRAGON AWAKENS!

Enemy Health: 60   |   Your Health: 90 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 1
You strike the enemy for 20 damage!
The enemy hits you for 15 damage!

Enemy Health: 40   |   Your Health: 75 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 1
You strike the enemy for 20 damage!
The enemy hits you for 15 damage!
```



```
Enemy Health: 20   |   Your Health: 60 / 100
[1] Attack  [2] Defend
[3] Use Potion  [4] Run
Enter choice: 1
You strike the enemy for 20 damage!
The enemy has been defeated!

========================================
                VICTORY!
========================================
CONGRATULATIONS, Vladimir!
You defeated the dragon and escaped!
Final Gold: 50
Remaining Health: 60
YOU ARE A HERO!
```

Assessment Rubric

Criterion	Points
Correct procedures (CALL/RET)	25
Conditional selection and branching	20
User input and validation	15
Correct logic and output	15
ASCII art and interface creativity	10
Looping and menu navigation	10
Readability and comments	5
TOTAL	100

Required Student Deliverables

Submit the .asm source file, screenshots of successful execution and invalid-input handling, a short demonstration video, and a brief explanation of implemented procedures.

Open-Ended Questions

1. What is the purpose of procedures in your application?

Procedures are used to divide the program into smaller and organized tasks. In this game, procedures such as display_status, display_inventory, battle_enemy, and use_potion each perform a specific function, making the code easier to read and manage. They also prevent code repetition because the same functionality can be called whenever needed. This helps simplify debugging and updating the game. The overall program becomes more structured and maintainable.

2. How do CALL and RET control program execution?

The CALL instruction transfers execution to a procedure while automatically saving the return address on the stack. For example, the main program uses CALL display_main_menu and CALL battle_enemy to execute specific tasks. After the procedure finishes, the RET instruction retrieves the saved return address and returns execution to the instruction immediately following the CALL. This allows different parts of the program to reuse the same procedures without duplicating code.

3. Which conditional jump instructions did you use, and why?

The program uses conditional jump instructions such as JE, JNE, JG, JGE, JL, and JLE. These instructions are used after CMP to make decisions based on user input and game conditions. For example, JE is used to select menu options, JLE checks if health reaches zero, and JG verifies that values remain above zero. These jumps control character

selection, combat outcomes, room progression, and input validation. They allow the game to follow different paths depending on the situation.

4. How does your program handle invalid user input?

The program validates user input using the read_choice procedure. It checks whether the entered character is between '1' and '9' and ensures only a single digit is entered. If the input is invalid, read_choice returns 0, causing error messages such as err_menu or err_class to be displayed. The program then loops back and asks the player for another input. This prevents unexpected values from affecting the game's execution.

5. What would be harder to maintain without procedures?

Without procedures, large sections of code such as combat, inventory display, and status display would have to be repeated throughout the program. Any changes would need to be made in multiple locations, increasing the chance of errors. For example, updating the battle system would require modifying every copied combat section instead of only the battle_enemy procedure. The code would become longer, harder to debug, and more difficult to understand. Procedures keep the program organized and easy to maintain.

Application-Specific Open-Ended Questions

Answer all questions for the application you developed. Write 3–5 complete sentences per question, refer to specific procedures or instructions from your own source code, and provide examples where requested.

Machine Problem 1 — Zodiac Horoscope and Fortune Teller

1. Explain how your determine_zodiac procedure identifies the correct sign from the birth month and day. Describe at least two conditional comparisons.

The procedure multiplies the month by 100 and adds the day to create a combined number. It uses cmp eax, 119 with jle to catch Capricorn, and cmp eax, 218 with jle to accurately identify all Aquarius signs. The number represents a MMDD format which triggers if the value is within a certain range (aka date format). For example, Capricorn is <= 119 or >= 1222.

2. How does your program handle zodiac date ranges that cross two months, such as Capricorn? Explain the branching logic.

The program converts the date to an MMDD format integer. Because Capricorn spans two years, it checks if the value is <= 119 (January 1-19) OR >= 1222 (December 22-31). By placing these two specific comparison branches at the very beginning, it successfully traps both ends of the Capricorn range before checking the other rest of the signs.

3. Why did you divide input collection, date validation, zodiac identification, and horoscope display into separate procedures?

The input collection does not ahve its own dedicated procedure like the rest that was mentioned. Input collection remained inline to avoid argument passing and to reduce code line usage. Date validation and zodiac identification were isolated into separate procedures to contain complex branching logic, simplifying debugging. The horoscope display was made to manage massive ASCII strings, keeping the main loop clean and ensuring the overall code remains highly organized and readable.

4. Describe how your program rejects invalid dates such as February 30 and an invalid month such as 13.

The validateDate procedure uses conditional jumps to reject months above 12. It then checks month values to cap February at 28 days, which DOES NOT consider leap years wherein February has 29 days.

5. How does the program select a horoscope category after the user enters a menu choice? Identify the conditional jump instructions used.

The program compares the user's menu choice using the CMP instruction. It then applies JE (Jump if Equal) instructions to branch directly to the correct category label for printing.

6. How did you implement the zodiac ASCII art? Explain how the correct design is selected and displayed.

The program uses conditional JE jumps to check the zodiac_id. Each jump targets a label that loads the corresponding art's address and length into registers, then calls printString.

7. What happens when a user enters an invalid menu option? Explain how control returns to the menu.

When an invalid menu option is entered, the program skips the valid comparisons, prints an error message, and uses a JMP instruction to loop directly back to the start of the main menu to try again.

8. If you added a new horoscope category, which procedures and conditional branches would you modify? Explain why.

I would add a new CMP and JE branch in the main menu loop to catch the new option. I would also create a new procedure label to print the new category's prediction; basically the same branching and procedure usage as the rest of the others.

Machine Problem 2 — Dungeon Adventure and Survival Game

1. Explain how initialize_character assigns different health, attack, and potion values for Warrior, Mage, and Archer.

The initialize_character procedure checks the value stored in class_id using CMP and conditional jumps. Warrior receives 100 HP, 20 Attack, and 2 Potions, Mage receives 70 HP,

30 Attack, and 3 Potions, while Archer receives 85 HP, 25 Attack, and 2 Potions. The procedure also stores the appropriate class name pointer and initializes game data such as gold, room number, and enemy health.

2. Describe the procedures and conditional jumps used when the player chooses Attack, Defend, Use Potion, or Run.

Inside battle_enemy, the player's choice is compared using CMP and selected through JE jumps. Choosing Attack goes to .attack, Defend goes to .defend_act, Use Potion calls use_potion, and Run goes to .run_act. Each action performs its specific logic before continuing or ending the battle.

3. How does your battle_enemy procedure determine whether the player or enemy has been defeated? Explain the comparisons.

After an attack, the enemy's health is reduced and compared against zero using CMP and JLE. If the enemy health is zero or below, execution jumps to .enemy_dead and returns victory. During retaliation, the player's health is reduced and compared against zero, and if it reaches zero, execution jumps to .player_dead, indicating defeat.

4. Explain how you prevent health from exceeding the maximum and prevent potion use when the inventory is empty.

The use_potion procedure first checks if the number of potions is greater than zero using CMP and JLE. If no potions remain, it displays msg_no_potion and exits. When healing, the player's health is compared to max_hp; if it exceeds the maximum, the value is replaced with max_hp to prevent overhealing.

5. How is the current dungeon room tracked, and what conditions allow the player to advance to the next room?

The current room is stored in the variable room. After defeating the goblin, the program sets room to 2, and after completing the treasure chamber, it sets room to 3. These updates allow the player to progress through the dungeon one room at a time.

6. What is the purpose of separating display_status, display_inventory, and battle_enemy into different procedures?

Separating these features into individual procedures keeps the program organized and modular. display_status handles character information, display_inventory manages

inventory output, and battle_enemy contains all combat logic. This makes the code easier to understand, test, and modify.

7. How does your program preserve important register values across procedures and Linux system calls? Give one example.

The program preserves registers by using instructions such as PUSH, POP, and PUSHAD/POPAD. For example, the print_string procedure uses PUSHAD before performing the Linux write system call and POPAD afterward to restore all general-purpose registers. This prevents important values from being accidentally overwritten.

8. If you introduced a fourth room with a stronger enemy, what program data, procedures, and conditions would need to change?

A new room variable value, enemy health variable, and enemy artwork would need to be added in the data and BSS sections. The explore_dungeon procedure would need a new room branch, and display_status would need to show the updated room count, such as "of 4" instead of "of 3". Additional conditions would also be added to advance from Room 3 to Room 4 and to handle the new enemy battle.

Submission Requirement for Written Responses

Include a separate “Open-Ended Questions” section in your submission. Answer the five general questions and the eight questions for your assigned machine problem. Answers must reflect your actual program implementation; unsupported or generic explanations will not receive full credit.

Instructor Guidance

Assign Zodiac Horoscope first to focus on procedures, branching, and input validation. Assign Dungeon Adventure afterward for arithmetic, game state, repeated decisions, and interacting procedures.